

Online Appendix to

Perfectly Still

Stability and Policy Response Capacity

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This appendix contains proofs, scope applications, sample construction, estimation details, and additional inference for the paper. The sections follow the notation and target definitions used in the main text.

Appendix A. Proofs

A.1. Proof of Theorem 1

PROOF. For each unit, the four values of $(I_{it}, I_{i,t+1})$ give

$$(A1) \quad I_{i,t+1}B_{i,t+1} - I_{it}B_{it} = (1 - I_{it})I_{i,t+1}B_{i,t+1} - I_{it}(1 - I_{i,t+1})B_{it}$$

$$(A2) \quad + I_{it}I_{i,t+1}(B_{i,t+1} - B_{it}).$$

Multiplying by ω_i and summing over units yields equation (6). $\square$

A.2. Proof of Proposition 1

PROOF. For a finite action set, evaluating the objective at every $k(a)$ gives equation (4). Population randomization with shares $p_a \geq 0$ and $\sum_a p_a = 1$ generates $\sum_a p_a k(a)$, whose range is $\text{conv}(\mathcal{K})$. Both $\mathcal{K}$ and $\text{conv}(\mathcal{K})$ are contained in $\text{span}(\mathcal{K})$, so minimizing the zero-cost quadratic loss over the span gives a weakly lower value. Equality holds exactly when the set of closest points in the span intersects the relevant attainable set. $\square$

A.3. Proof of Theorem 4

PROOF. Both systems follow $\delta s_{t+1} = A\delta s_t$ under the baseline policy, so their passive trajectories coincide and converge because $\rho(A) < 1$. For $B_0 = 0$, every block in equation (20) is zero. For $B_1 = I_n$, the final block is $CB_1 = C$, which has row rank q . The full horizon matrix therefore has row rank q . $\square$

A.4. Proof of Theorem 5

PROOF. Local Lipschitz continuity and $M_H(s^*) = 0$ give $\|M_H(s_t)\|_F \leq L\|s_t - s^*\|$. Exponential attraction gives equation (21). Solving $LK\lambda^t\|s_0 - s^*\| < \kappa$ for t gives the capacity-floor date. $\square$

A.5. Proof of Theorem 2

PROOF. For any bundle S , the condition $W \subseteq V + \text{Im } D_S$ is equivalent to $\pi_V(W) \subseteq \text{span}\{\pi_V(d_\ell) : \ell \in S\}$. Taking S as the full dictionary proves the existence statement. If S is feasible, then

$$(A3) \quad d(W, V) = \dim \pi_V(W) \leq \dim \text{span}\{\pi_V(d_\ell) : \ell \in S\} \leq |S|.$$

The lower bound is attained exactly when a set of $d(W, V)$ projected dictionary vectors contains $\pi_V(W)$ in its span. For unrestricted additions, choose vectors $w_1, \dots, w_d \in W$

whose quotient classes form a basis of $\pi_V(W)$ and activate them at unit cost. Their span with V contains W , so both minima equal d . $\square$

A.6. Proof of Theorem 3

PROOF. The optimizer in equation (16) is $P_V^Q Z$, so the realized residual is $R_V^Q Z$. A Q -orthogonal projector satisfies $(R_V^Q)' Q = Q R_V^Q$ and $(R_V^Q)^2 = R_V^Q$. Taking expectations gives

$$(A4) \quad L(V) = \frac{1}{2} \mathbb{E}[Z'(R_V^Q)' Q R_V^Q Z] = \frac{1}{2} \text{tr}(Q R_V^Q \Omega).$$

When $\tilde{d} \neq 0$, the new component added to the fitted target change is the Q -orthogonal projection of $R_V^Q Z$ onto $\text{span}\{\tilde{d}\}$. Its squared Q -norm is

$$(A5) \quad \frac{(\tilde{d}' Q R_V^Q Z)^2}{\tilde{d}' Q \tilde{d}}.$$

The projection theorem makes one half of its expectation equal to the reduction in equation (16). Substitution of $\Omega = \mathbb{E}[ZZ']$ gives equation (18). When $\tilde{d} = 0$, the addition lies in V and leaves the feasible target-change set unchanged. $\square$

A.7. Proof of Proposition 2

PROOF. Write $x_1 = x(1)$ and $x_0 = x(0)$, with $x_1 \leq x_0$. Since $\min\{x, \ell\} = \int_0^\ell \mathbf{1}\{z \leq x\} dz$, equation (22) gives

$$(A6) \quad K(\ell) = - \int_0^\ell \mathbf{1}\{x_1 < z \leq x_0\} dz.$$

Subtracting the integrals at ℓ_1 and ℓ_0 gives the interval length in equation (23). Applying the identity unit by unit and taking expectations gives equation (24). For incentive-dependent caps, subtract unrestricted use and write

$$(A7) \quad \begin{aligned} & \min\{X(1), L_1\} - X(1) - [\min\{X(0), L_0\} - X(0)] \\ &= (X(0) - L_0)_+ - (X(1) - L_1)_+. \end{aligned}$$

Adding and subtracting $(X(1) - L_0)_+$ gives equation (25) because $(X(0) - L_0)_+ - (X(1) - L_0)_+$ equals the displayed interval length. $\square$

A.8. Proof of Proposition 3

PROOF. Write $c = c(a)$. Conditional on v , a household below the first-period cutoff enters Phase 2 without early adoption and purchases when $v \geq p_2$. A household above the

cutoff enters with early adoption and purchases when $g \geq p_2 - v$. Hence

$$(A8) \quad K_2(a, p_2) = \int_{-\infty}^c \mathbf{1}\{v \geq p_2\} f_v(v) dv$$

$$(A9) \quad + \int_c^{\infty} \Pr\{g \geq p_2 - v \mid v\} f_v(v) dv.$$

Leibniz's rule gives the derivative with respect to c . Multiplication by $dc/da = -1$ yields equation (29). $\square$

A.9. Proof of Proposition 4

PROOF. Differentiate equation (32). Equation (30) gives the derivative of current adoption, and equation (29) gives the derivative of later capacity. Equating marginal benefit and marginal cost gives equation (33). $\square$

A.10. Proof of Theorem 6

PROOF. Iteration of the second transition in equation (34) gives $r_t = (1 - \alpha)^t r_0$. The population identity gives $x_t = 1 - e - r_t$, and equation (35) gives $K_t = \alpha r_t$. Since $\alpha \in (0, 1)$, all three limits in equation (36) follow. At the limit, another campaign acts on a reachable pool of size zero. Its capacity contrast is therefore zero. If $e > 0$, every column generated by the same campaign has zero projection onto changes in the excluded group, so additions within that span leave the gap e unchanged. $\square$

A.11. Proof of Proposition 5

PROOF. Sequential assignment and consistency give

$$(A10) \quad \mathbb{E}[Y_{t+1+h}(a, b) \mid H_t = h_t] = \int \mathbb{E}[Y_{t+1+h} \mid H_t = h_t, A_t = a, L_{t+1} = \ell, B_{t+1} = b] dF(\ell \mid h_t, a).$$

The smooth-support and envelope conditions permit differentiation with respect to b under the integral, which gives equation (40). A second application of the maintained interchange conditions with respect to a identifies $\theta_h(a, h_t)$. $\square$

A.12. Proof of Proposition 9

PROOF. Joint asymptotic normality and covariance consistency imply convergence of the studentized contrast vector to its Gaussian law. The definition of $c_{1-\alpha}$ gives simultaneous coverage of every component. On that simultaneous-coverage event, the interval chosen by any selector with range in the finite family also covers its contrast. $\square$

A.13. Proof of Proposition 10

PROOF. The availability restriction gives $K_2(a) = \mathbb{E}[Y_2(a, 1)]$. Conditional random assignment gives $\mathbb{E}[Y_2(a, 1) \mid G] = \mathbb{E}[Y_2 \mid A = a, G]$. Subtracting the expression for a' and averaging over the included blocks gives equation (50). $\square$

A.14. Proof of Proposition 7

PROOF. Conditional random assignment and consistency give $\mathbb{E}[Y_h(d, b) \mid G] = \mathbb{E}[Y_h \mid D = d, B = b, G]$ for each factorial cell. Substitute the four observed conditional means into equation (44), take the two within-configuration intervention contrasts, and average over the included strata. This yields equation (45). $\square$

A.15. Proof of Proposition 8

PROOF. Proposition 7 identifies ΔK_h for every $h \in \{h^*\} \cup \mathcal{C}$. Apply the weights in equation (47) to these identified cell-mean contrasts. If $\widehat{\theta}$ stacks the corresponding interaction coefficients and a stacks the weights, the estimator is $a'\widehat{\theta}$ and its covariance is $a' \text{Var}(\widehat{\theta})a$. $\square$

Appendix B. Scope Applications

B.1. Sequential bed-net subsidy exposure

The application uses the design and public replication data of Dupas (2014). The experiment introduced Olyset long-lasting insecticide-treated bed nets to 1,120 households in six rural areas of Busia District, Kenya. In Phase 1, households received vouchers with randomized subsidy levels and had three months to redeem them. Four areas entered Phase 2 one year later. All 599 households in those areas received an unannounced voucher offering another Olyset net for 150 Kenyan shillings. The product was unavailable through local retailers during the experiment, while other bed nets remained available. Purchase from the Phase 2 voucher therefore gives the response to a defined later offer.

Define $H_i = \mathbf{1}\{P_{1i} \leq 50\}$, where P_{1i} is the randomized Phase 1 price. Let G_i be the share of sampled households within 500 meters whose assigned price also lies at or below 50 shillings. The response targets are D_{1i} , purchase in Phase 1, and D_{2i} , purchase from the Phase 2 offer. For households with Phase 2 outcomes and valid coordinates, the estimated system is

$$(A11) \quad \begin{bmatrix} D_{1i} \\ D_{2i} \end{bmatrix} = \alpha_{g(i)} + M^A \begin{bmatrix} H_i \\ G_i \end{bmatrix} + \Gamma Z_i + \varepsilon_i,$$

where $\alpha_{g(i)}$ contains experimental-area effects and Z_i contains the randomized voucher-recipient and marketing-pitch assignments. The first column of M^A is the own-assignment response vector. The second column is the linear exposure-response coefficient for randomized neighborhood assignment.

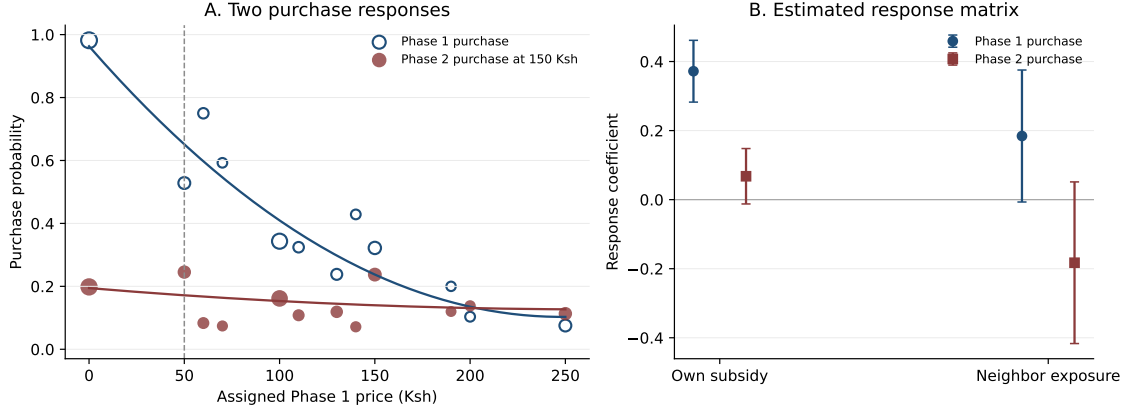


FIGURE A1. Randomized subsidy history and purchase response

Notes. Panel A plots purchase rates by assigned Phase 1 price. Marker area reflects the number of households at each price and lines are observation-weighted quadratic fits. Every Phase 2 household faced a price of 150 Kenyan shillings. Panel B reports the rows of $\hat{M}^A$ from equation (A11); bars are 95 percent intervals from one-kilometer spatial HAC standard errors.

In the 599-household Phase 2 sample, the own high-subsidy coefficient is 0.372 for Phase 1 purchase and 0.0718 for Phase 2 purchase. The within-area randomization values are 0.0001 and 0.0742. Table A1 uses the 584 households with valid coordinates. The covariance estimator stacks the two equation scores and applies a Bartlett spatial kernel to household pairs within one kilometer (Conley 1999).

TABLE A1. Current adoption and later offer response

	Phase 1 purchase	Phase 2 purchase
Own high subsidy	0.372 (0.046)	0.068 (0.041)
Neighbor high-subsidy share	0.184 (0.098)	-0.183 (0.119)
Singular values	0.416, 0.193	
Rank-zero Wald test	69.03 ($p < 0.001$)	
Rank-one determinant test	-0.080 (0.047), $p = 0.090$	
Lower-rank-uniform rank-one test	21.86, $p = 0.077$	
Observations	584	

Notes. Each column of the response matrix corresponds to an assigned policy exposure, and each row corresponds to a purchase outcome. High subsidy means a Phase 1 assigned price at or below 50 Kenyan shillings. Neighbor exposure is the share of households within 500 meters assigned a high subsidy. Regressions include experimental-area effects and the randomized voucher-recipient and marketing-pitch assignments. Parentheses report spatial HAC standard errors with a one-kilometer Bartlett kernel. The rank-zero row is a four-degree-of-freedom Wald test. The determinant row uses a delta approximation around a regular rank-one matrix. The lower-rank-uniform row applies the Chen–Fang CFA two-step test to the spatial-HAC coefficient law. Targets are measured as purchase probabilities and instruments as unit changes in assignment exposure.

The estimated response matrix is

$$(A12) \quad \widehat{M}^A = \begin{bmatrix} 0.372 & 0.184 \\ 0.068 & -0.183 \end{bmatrix}.$$

The spatial standard errors for the own column are 0.046 and 0.041. The singular values are 0.416 and 0.193. The rank-zero Wald statistic is 69.03, with four degrees of freedom and $p < 0.001$. The determinant is -0.080 with a delta standard error of 0.047. The Chen–Fang two-step statistic is 21.86 with a Gaussian bootstrap value of 0.077. The rank-zero result identifies one response direction; inference for the second direction crosses the 5 percent threshold.

Among 599 households, 282 purchased from neither offer, 223 purchased only in Phase 1, 38 purchased only in Phase 2, and 56 purchased in both phases. Random assignment identifies the total own-history effect in equation (50). The 7.2-percentage-point later estimate combines experience, ownership, and other pathways through which early assignment changes response to the later offer. The single Phase 2 price identifies this con-

trast at 150 shillings. A later randomized price schedule would identify the corresponding demand curve.

B.2. Repeated reach in StayWell

The StayWell at Home Phase II program followed 266 adults for 60 days (Arévalo Avalos et al. 2024). Each day assigned a behavioral-activation tip, a coping-skills tip, a social-support tip, or a tip-control condition with probability one quarter. A mood-rating prompt followed three hours later across the four categories. The public OSF file records message assignment and receipt of a mood response. The repeated-reach outcome is a participant’s first observed reply to that common prompt.

The accounting sample contains the 177 participants with at least 50 recorded decision dates and uses their first 50 prompts. The first prompt reaches 145 participants, or 81.9 percent of the cohort. The second and third prompts add 11 and 4 participants. Three further first responses occur by prompt 11. Prompts 12 through 50 add zero observed first responses, while 13 participants remain in the gap. Cumulative reach is 92.7 percent and the residual gap is 7.3 percent, with an exact 95 percent interval from 4.0 to 12.2 percent.

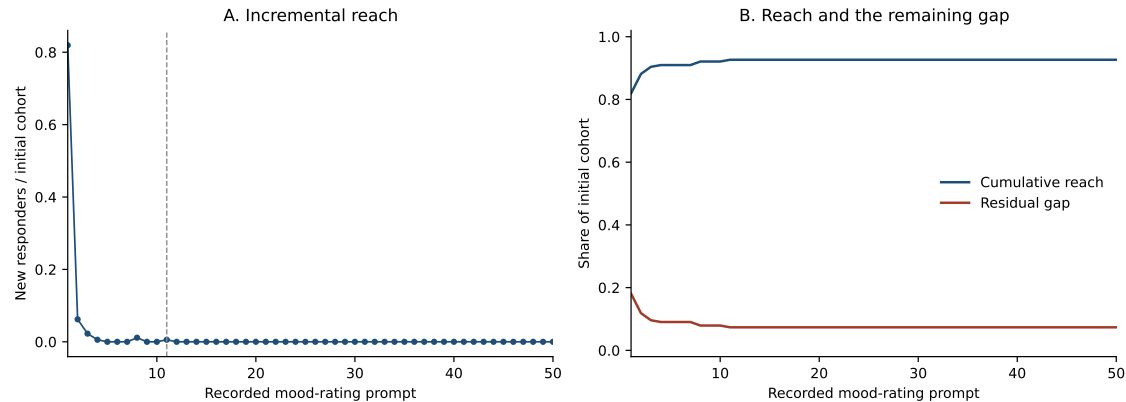


FIGURE A2. Repeated prompt reach and the residual response gap

Notes. The sample contains 177 StayWell Phase II participants with at least 50 recorded decision dates. Incremental reach is the share of this cohort supplying its first observed reply to the daily mood-rating prompt at each decision. Cumulative reach and the residual gap sum to one.

TABLE A2. Repeated reach of the StayWell mood-rating prompt

Prompt number	1	2	3	11	20	50
New responders	145	11	4	1	0	0
Cumulative reach	0.819	0.881	0.904	0.927	0.927	0.927
Residual gap	0.181	0.119	0.096	0.073	0.073	0.073

Notes. The sample contains 177 Phase II participants with at least 50 recorded decision dates. New responders supply their first observed reply to the daily mood-rating prompt at the listed decision. The prompt was common across randomized CBT-tip categories, so the table is repeated-reach accounting. The residual gap after 50 prompts is 13 participants, or 0.073 of the cohort.

The sequence supplies the observed incremental reach path in equation (37). The common prompt appears in every randomized tip category, so its effect lies outside the experiment’s treatment contrast. The fixed cohort also conditions on at least 50 recorded decision dates. Exclusion from the prompt’s reach, low response probability, attrition, and missing measurement remain combined in the residual gap.

Appendix C. myBPmyLife Micro-Randomized Trial

C.1. Decision sample

The analysis uses the decision-level data, core-cohort identifiers, baseline step counts, and codebook released with Golbus et al. (2026).

The core identifier file selects 287 participants and 187,517 recorded randomization points, matching the source article. The active phase contains 186,893 points with a recorded dice roll, spanning relative enrollment days 8 through 180. A successful activity roll occurs at 12.52 percent of these points and a successful diet roll at 12.36 percent. The corresponding design probabilities are 12.5 percent each. The codebook defines `IsPresent_DeviceDataPoints_30minBefore` as recorded app use during the 30 minutes before assignment and `IsPresent_DeviceDataPoints_60minAfter` as recorded app use during the next 60 minutes. The count endpoints record app events, searches, and low-salt choices over the same windows.

C.2. History construction and estimator

Within participant, observations are ordered by relative enrollment day, time of day, preference timestamp row, and dice-roll row. For each window $w \in \{4, 12, 28, 56, 112\}$, the analysis sums successful activity and diet assignments across the preceding w recorded decision points and centers the sum at $0.25w$. A row enters a window specification after

all w preceding assignments are observed. The resulting samples contain 185,745, 183,449, 178,857, 170,821, and 154,776 decision points.

Equation (42) is estimated by least squares after absorbing participant, relative-day, and time-of-day effects. The explicit regressors are centered current diet assignment, centered current activity assignment, centered randomized history, both current-assignment-by-history products, the endpoint measured in the preceding 30 minutes, both current-assignment-by-pre-state products, and centered linear and quadratic relative-day terms interacted with both current assignments. Participant-cluster CR1 covariance estimates give pointwise standard errors. Gaussian cluster-multiplier maximum statistics from the joint score covariance give the simultaneous critical values and adjusted values.

TABLE A3. Passive app-use transition at control-draw decision points

Quantity	Estimate	Cluster SE
App use after pre-state 0	4.22 pp	0.25 pp
App use after pre-state 1	19.76 pp	1.03 pp
Transition slope	0.155	0.010
Implied two-state fixed point	5.00 pp	0.31 pp

Notes. The sample contains 140,397 active-phase decision points assigned to the 0.75 control condition. The pre-state records app use during the preceding 30 minutes, and the outcome records app use during the next 60 minutes. The first two rows come from a linear transition regression with participant-clustered standard errors. The transition slope is their difference. Iterating the corresponding two-state kernel gives the displayed fixed point.

TABLE A4. Randomized-history coefficient family

Outcome	Days	Diet history (SE)	Adjusted p	Activity history (SE)	Adjusted p
App use	1	-0.3041 (0.2527)	0.9896	-0.3552 (0.2160)	0.8489
App use	3	-0.1800 (0.1289)	0.9539	-0.1037 (0.1264)	1.0000
App use	7	-0.2745 (0.0947)	0.0878	-0.1278 (0.0830)	0.9040
App use	14	-0.2193 (0.0745)	0.0766	-0.0384 (0.0617)	1.0000
App use	28	-0.0870 (0.0499)	0.7856	0.0209 (0.0486)	1.0000
App events	1	-0.0333 (0.0325)	0.9985	-0.0382 (0.0257)	0.9245
App events	3	-0.0253 (0.0173)	0.9346	-0.0158 (0.0141)	0.9953
App events	7	-0.0330 (0.0129)	0.2069	-0.0145 (0.0091)	0.8814
App events	14	-0.0264 (0.0092)	0.0929	-0.0003 (0.0070)	1.0000
App events	28	-0.0100 (0.0054)	0.7124	0.0027 (0.0053)	1.0000
Searches	1	-0.0029 (0.0058)	1.0000	-0.0051 (0.0049)	0.9985
Searches	3	-0.0047 (0.0034)	0.9522	-0.0029 (0.0025)	0.9942
Searches	7	-0.0050 (0.0024)	0.5295	-0.0016 (0.0016)	0.9984
Searches	14	-0.0034 (0.0017)	0.6226	0.0001 (0.0012)	1.0000
Searches	28	-0.0010 (0.0011)	0.9997	0.0004 (0.0009)	1.0000
Low-salt choices	1	-0.0062 (0.0065)	0.9995	-0.0045 (0.0044)	0.9983
Low-salt choices	3	-0.0034 (0.0034)	0.9988	-0.0019 (0.0028)	1.0000
Low-salt choices	7	-0.0061 (0.0026)	0.3028	-0.0017 (0.0017)	0.9994
Low-salt choices	14	-0.0047 (0.0018)	0.1649	0.0004 (0.0013)	1.0000
Low-salt choices	28	-0.0021 (0.0011)	0.5746	0.0008 (0.0009)	0.9998

Notes. Entries are current-notification-by-randomized-history coefficients with participant-clustered standard errors. App-use coefficients are percentage points per additional past assignment; the three count outcomes use counts per additional past assignment. Adjusted values use the maximum absolute studentized statistic over five windows, four outcomes, and two notification types. Every specification includes the controls and fixed effects listed in Table 1.

The five-coefficient app-use family has a 95 percent simultaneous critical value of 2.493. The 40-coefficient family has a critical value of 3.090. For the 14-day history, the app-use diet interaction equals -0.00219 with standard error 0.00074, displayed-family $p = 0.0136$, and 40-coefficient-family $p = 0.0766$. Its 10th and 90th history percentiles are 10 and 18 assignments. For the 7-day history, the interaction equals -0.00274 with standard error 0.00095 and corresponding adjusted values of 0.0159 and 0.0878. Its history percentiles are 4 and 10 assignments.

A future-history specification constructs a centered sum from the next 28 randomization points and applies the same current-assignment, pre-state, and relative-time terms. The diet-by-future-history coefficient equals 0.00119 with standard error 0.00094 and $p = 0.206$; the activity coefficient equals 0.00131 with standard error 0.00090 and $p = 0.150$.

Appendix D. Digital-Use Experiment and Estimation

D.1. Treatment assignment and sample

The application uses version 2.0 of the Harvard Dataverse deposit for Allcott et al. (2022). The experiment is registered as AEARCTR-0005796. The sample contains 1,933 randomized participants, and the codebook supplies survey and phone-dashboard variable definitions.

Bonus assignment is `S3_Bonus`. Limit configuration is `S2_LimitType`: zero denotes limit control, values one through four correspond to snooze delays of 0, 2, 5, and 20 minutes, and value five denotes a limit without snooze. The binary limit indicator equals one for values one through five. The four bonus-by-limit cells contain 589, 194, 865, and 285 participants. Average daily FITSBY use in period h is `PD_Ph_UsageFITSBY`. Period 1 supplies baseline use, period 3 is the bonus-payment period, and limit assignment applies from period 2. Randomization forms eight baseline strata, assigns bonus status within strata, and assigns blocker status and snooze duration within the corresponding stratum-by-bonus cells.

D.2. Factorial regression

For periods $h = 2, \dots, 5$, the analysis estimates

$$(A13) \quad Y_{ih} = \alpha_h + \beta_h B_i + \delta_h D_i + \theta_h B_i D_i + \gamma_h Y_{i1} + \lambda_{g(i),h} + \varepsilon_{ih},$$

where B_i is bonus assignment, D_i is limit assignment, and $\lambda_{g(i),h}$ contains the reported randomization-stratum indicators. Equation (45) gives $K_h(0) = \beta_h$, $K_h(1) = \beta_h + \theta_h$, and $\Delta K_h = \theta_h$. HC2 standard errors provide horizon-specific inference. The joint periods 2–5 test uses the cross-horizon covariance of the four interaction influence functions. The payment-overlap contrast applies weights $(-1/3, 1, -1/3, -1/3)$ to $(\widehat{\theta}_2, \widehat{\theta}_3, \widehat{\theta}_4, \widehat{\theta}_5)$. The simultaneous period family applies the four rows of $(4/3)I_4 - (1/3)\mathbf{1}\mathbf{1}'$ to the HC2 contrast covariance. The period-1 balance regression removes Y_{i1} from the covariate set.

The weekly specification uses `PD_WeeklyUsageFITSBY_w` for weeks $w = 4, \dots, 15$ and week 3 as the baseline outcome. It estimates equation (A13) on the common 1,903-person sample. Each of ten contrasts assigns weight $1/3$ to one consecutive three-week window and weight $-1/9$ to the other nine weeks. The payment-overlap contrast selects

weeks 7–9. Cross-week HC2 influence functions supply the covariance for the maximum statistic.

The raw response matrix stacks β_h and δ_h for periods 2–5 and divides each row by the sample standard deviation of its outcome. Cross-horizon influence functions form the covariance matrix of the eight response coefficients. The bonus exposure schedule is $s_B = (0, 1, 0, 0)'$ and the limit access schedule is $s_D = (1, 1, 1, 1)'$. Let $S = (s_B, s_D)$ and $P_S = S(S'S)^{-1}S'$. The schedule-adjusted matrix is $M^\perp = (I - P_S)M$. Applying the same linear transformation to every influence function supplies the covariance of $M^\perp$. The raw singular values are 0.821 and 0.293, while the adjusted singular values are 0.094 and 0.024. The lower-rank-uniform test applies equation (51) with $\kappa_n = n^{-1/4}$ to the transformed HC2 covariance. Its value for adjusted rank at most one is 0.066. A Gaussian max- t family covers the six configuration-specific period-3 responses, and a second family covers the five response comparisons with limit control.

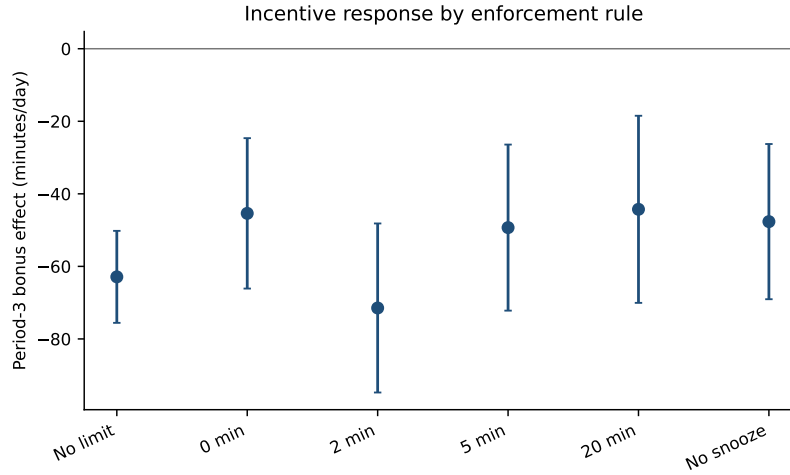


FIGURE A3. Period-3 incentive response by randomized enforcement rule

Notes. Markers report the period-3 bonus effect within the limit-control group and the five randomized limit configurations. Bars give 95 percent Gaussian max- t simultaneous intervals based on the HC2 contrast covariance.

D.3. Randomization inference and planner calculation

Conditional randomization first permutes bonus assignment within each recorded stratum and then permutes detailed limit labels within the corresponding stratum-by-bonus cells, preserving the sequential treatment counts. Each draw keeps outcomes fixed and re-estimates the saturated factorial regression and HC2 standard error. The studentized statistic supplies randomization inference for a weak null of zero average factorial interaction under treatment-effect heterogeneity (Wu and Ding 2021). The period-3 two-sided value is 0.0694. Applying the four one-period-versus-rest contrasts to the common periods

2–5 sample gives 0.0059 for the payment contrast and 0.0216 after maximum-studentized family adjustment.

For the targeting calculation, Q divides each horizon by its outcome variance in the control assignment cell. A declared common daily use-reduction target r gives $z(r) = (-r, -r, -r, -r)'$. The four feasible target changes are $k(0, 0) = 0$, $k(1, 0) = \widehat{\beta}$, $k(0, 1) = \widehat{\delta}$, and $k(1, 1) = \widehat{\beta} + \widehat{\delta} + \widehat{\theta}$. Equation (4) evaluates these four points. Over targets from zero through 80 minutes, the estimated zero-cost boundaries are 11.61 and 41.53 minutes. At targets of 15, 30, and 50 minutes, the selected actions are limit, limit, and the combined assignment, with joint-coefficient selection probabilities 0.998, 1.000, and 0.999. Survey responses imply a mean expected agency bonus outlay of \$76.42. With objective $\lambda L(a) + C(a)$ and limit implementation cost c_L , a 30-minute target selects the limit when $c_L < 0.184\lambda$ relative to control and when $c_L < 76.42 + 0.109\lambda$ relative to the bonus. At a 50-minute target, the combined assignment is selected for $\lambda > 929.9$ under the observed bonus outlay and zero recorded limit outlay.

Appendix E. StayWell Repeated-Reach Accounting

The analysis uses the Phase II workbook associated with Arévalo Avalos et al. (2024); the study is registered as NCT04473599. Phase II assigns each participant daily among behavioral activation, coping skills, social support, and tip control with probability one quarter for each category, followed three hours later by a mood-rating prompt. Removing one copy of 1,192 exact duplicate rows in the Random arm and 49 rows with message codes outside the four recorded categories yields 14,258 decision rows for 266 participants.

The repeated-reach sample fixes exposure support by retaining 177 participants with at least 50 recorded decision dates and using the first 50 dates for each participant. A first response occurs on the first date with an observed mood reply. Incremental capacity at date t is the number of first responses at t divided by 177. Cumulative reach sums this path, and the residual gap is one minus cumulative reach. The sequence contains 145 first responses at decision 1, 11 at decision 2, 4 at decision 3, 1 at decision 4, 2 at decision 8, and 1 at decision 11. The remaining 13 participants supply zero observed replies through decision 50. Exact binomial inversion gives a 95 percent interval of 0.040 to 0.122 for their cohort share.

Appendix F. Bed-Net Experiment and Estimation

F.1. Data and sample

The analysis uses the replication files released with Dupas (2014). Fieldwork ran in 2007–08. The household file contains the randomized Phase 1 price, purchase outcomes for

both phases, experimental-area indicators, marketing assignments, and neighborhood exposure variables used below.

The household file contains 1,120 rows. The variable `price` is the randomized Phase 1 price. The variables `purchasednet` and `purchasednet2` record redemption from the Phase 1 and Phase 2 shopkeeper logs. Phase 2 purchase is observed for 599 households in four experimental areas. Coordinates are available for 584 of those households. The variables `n_cheap500` and `n_total500` count nearby high-subsidy and sampled households within 500 meters. The analysis defines neighborhood exposure as their ratio and assigns zero when the neighborhood count is zero, matching the construction in the original code.

F.2. Estimation sample and response matrix

Let $Y_i = (D_{1i}, D_{2i})'$ and $T_i = (H_i, G_i)'$. Equation (A11) is estimated by least squares on the common 584-household sample. The control vector contains an intercept, three experimental-area indicators, assignment of the voucher to a male respondent or to both respondents, and the burden and cost marketing pitches. The coefficient array is arranged with outcomes in rows and policy exposures in columns.

For coordinates (ϕ_i, λ_i) in radians, pairwise distance is the haversine distance. With cutoff $c = 1$ kilometer, the kernel weight is $w_{ij} = \max\{1 - d_{ij}/c, 0\}$. If X is the equation design matrix, $Q = (X'X)^{-1}$, and e_{ir} is the residual for outcome r , the cross-equation covariance block is

$$(A14) \quad \widehat{V}_{rs} = \frac{N}{N-K} Q \left(\sum_i \sum_j w_{ij} x_i x_j' e_{ir} e_{js} \right) Q.$$

Stacking these blocks gives the covariance of the two-equation coefficient vector. This is the covariance used for Table A1 and the rank calculations.

The rank-zero statistic is $\text{vec}(\widehat{M})' \widehat{V}_M^{-1} \text{vec}(\widehat{M})$. For the determinant calculation, the coefficient order is $(m_{11}, m_{12}, m_{21}, m_{22})$ and the gradient is $(m_{22}, -m_{21}, -m_{12}, m_{11})'$. Substitution into equation (52) yields the reported standard error under the regular rank-one approximation. The lower-rank-uniform calculation uses the same coefficient covariance with $\kappa_n = 584^{-1/4} = 0.203$. The estimated lower rank is one, the statistic is 21.86, and the bootstrap p -value is 0.0766.

F.3. Randomization inference and transitions

The own-assignment randomization test uses all 599 households with Phase 2 outcomes. The high-subsidy indicator is permuted within each of the four experimental areas, preserving each area's treated count. Each draw residualizes the permuted assignment and

each outcome on area effects and the other randomized assignments. Two-sided randomization values compare the absolute observed coefficient with the permutation distribution.

The observed purchase histories contain 282 households with $(D_1, D_2) = (0, 0)$, 38 with $(0, 1)$, 223 with $(1, 0)$, and 56 with $(1, 1)$.

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